

CHAPTER THREE

PURCHASING

3.1. Meaning and objectives

Purchasing function comprises the essential activities associated with the acquisition of material used in the operation of an organization.

Because all organization requires supplies of materials, purchasing functions is common in almost all organizations.

Dear student, purchasing is both science and art

- There are two basic types of purchasing in business

➤ Purchasing for sale

➤ Purchasing for consumption/conversion

- Purchasing for sale is performed primarily by merchants.
- Purchasing managers who buy materials for consumption or conversion are called industrial buyers or purchasers.

This term includes buyers for

- Manufacturing firms
- Service businesses
- Institutions (school, hospitals, etc.)
- Government agencies

3.1.1 Objectives of purchasing

The objectives of purchasing can be viewed from three levels:

1. a very general managerial level
2. more specific operational level
3. a detail level

1. General managerial level:

From their prospective, relates to the five rights the managements expects the purchasing department to achieve:

- Of the right quality
- Of in the right quantity
- From the right supplier
- At the right time
- At the right price

A sixth factors implied in these terms includes the desired services necessary for optimum supply and utilization of the materials.

2. Functional level

1. To support company operations with an interrupted flow of materials
 - this is the most fundamental of all purchasing objectives
 - in a logical sense this is a key reason for the existence of the department
2. To buy competitively; buying competitively involves:
 - keeping abreast of the forces of supply and demand that regulate prices and availability of materials in the market place
 - Understanding of suppliers cost structure computed with an ability to help improve the cost structure.

Note. A buyer who pays significantly more than his/her competitor does for a given

Material is not buying competitively

3. To buy wisely: buying wisely involves,
 - A continued search for better values that yield the best combination of quality service and price. This intern involves coordination with users. In users' needs and reconciling users' needs with suppliers capabilities

Example: A firm that purchases high-grade bond paper for use in internal office communications that could be performed as well with less expensive softer stock paper usually in not buying wise.

Note. It is the combination of buying competitively and buying wisely that typically contribute most to the profitability of the firm.

4. To keep inventory investment and inventory losses at a practical minimum.

- Although maintaining a large inventory is one way to achieve objectives, it is also costly. Hence, the purchasing department job is to achieve a reasonable balance because the level of inventory required supporting operations and cost of carrying the inventory.
 - JIT (Just-in-time) production inventory system helps considerably in achieving their objective.
 - Though proper buying, handling, storing, if is also the department is responsible to minimize losses that occur.
5. To develop effective and reliable source of supply.
 - It involves the identification, investigation, selection and development of competent and responsive suppliers.
 - Progressive buyers tend increasingly to “buy suppliers’ as opposed to simply “buying products” which may lead to develop partnering arrangements/strategic alliance with the suppliers.
 6. To develop good relationship with supplier community and good continuing relationship with active suppliers.
 7. To achieve maximum integration with other departments of the firm.
 - understand material needs of user department
 - Support user department in actions like material standardization, forecasting future prices, performing more or by analysis.
 8. To hand the purchasing functions proactively. In a professional, cost effective manner.

Note. These objectives apply in principle to all categories of industrial buying

3.2. Purchasing policies and procedures.

Policy

- Policies are general statements, understanding, and guidelines in making operating decisions that channel actions toward achievement of the objectives.
- Purchasing policies are aids for purchasing decisions. *Purchasing* policy may be written or not
- Policies Defining Purchasing Authority
- It is a policy defining the placement of purchasing authority
- There are two approaches of placing purchasing function

➤ Centralization

➤ Decentralization

A. Centralized Purchase

Centralization exists when the entire purchasing function is made the responsibility of a single person.

This person is held accountable for performance of purchasing activities.

Advantage

- Duplication of effort and haphazard purchasing activities are minimized by the central coordination of all company purchasers.
- Quantity discounts are made possible by consolidating all company orders for the same and similar materials.
- A firm is able to develop and implement a unified purchasing policy, enabling it to speak with a single voice to its suppliers.
- Transportation savings can be realized by the consolidation of orders and delivery schedules.
- Develops purchase specialists
- Suppliers are able to offer better prices and better service because their sales personnel:
 - have fewer people to call on,
 - fewer shipments to make
 - fewer invoices to prepare
 - fewer financial records to

Line department manager do not have to spend time on purchasing. They can do not full time and effort to their basic responsibilities.

Fewer orders are processed for the same quantity of goods purchased, thereby reducing purchasing, receiving, inspection accounts payable and recordkeeping expenses.

- Responsibility for the performance of the purchasing function is fixed with a single department head, thereby facilitating mgt control.

Disadvantage

- Slow decision making
- May not satisfy local interest
- Does not spread risk

Factors affecting feasibility and desirability of centralization

1. Similarity of the classes of materials used in each of the depts. /plants. If a firm's plants (products) use entirely different materials, centralization of purchasing offers only minimal benefits.
But centralization offers benefits b/c of specialization when the various plants use similar materials similarity of materials usage depends on the line of products manufactured by a company.
2. Size of each individual plant purchasing dept. when a firm's individual plant purchasing dept. is not large in size, centralization would be advantageous.
3. Distance separating individual plants or geographic dispersion of plants the closer a firm's plants are situated geographically, the more feasible centralization becomes, conversely the wider the plants are dispersed, the more serious disadvantages of centralization become.

B. Decentralized purchase

Decentralization of purchasing occurs when personnel from other functional areas—operation, marketing, finance, HRM, etc.—decide unilaterally on sources of supply or negotiate with suppliers directly for major purchases.

Advantage

- Speed of operation there is a possibility of response more quickly to user's needs and time delay can be improved.
- Effective use of local resources/local interest
- Plant/dept. autonomy

Disadvantage

- Losing control
- Difficulty of obtaining discounts
- Duplication of effort

Factors that favor decentralized purchase

1. Single natural raw materials – knowledge of these materials is important
2. Technically-oriented firms that are heavily evolved in research
 - These firms some exceptions to complete centralization are always desirable.Many onetime purchases in the research design engineering and related depts... Can be handled more effectively by professional personnel in these depts...

In the research situation

- Scientists (researchers) must frequently discuss concepts with one or more suppliers before it is possible to select specific pieces of hardware.

Design

- Design engineers concerned with developing a prototype must frequently talk with the supplier at the completion of each developmental step.
3. Operation of multi-site institutional and manufacturing organs.
 4. Purchase of non-technical odds and ends

Example: Credit card and petty cash fund purchases Decentralizing through the use of these approaches can be a money saver.

Purchasing Procedure

Purchasing procedure/purchasing cycle/purchasing process comprises the following basic steps.

1. Recognize, define and describe the need (purchase requisition)
2. Verification of purchase requisition
3. Request for bid
4. Evaluation and selection of suppliers
5. Issuance of purchase order
6. Follow up and expediting
7. Receipt, and inspection of materials
8. Checking of invoices and bill payment
9. Completion of the records and files
10. Evaluation of the purchase process

1. Recognition, Definition and Description of need Purchase Requisition (PR)

- The need for purchase typically originates in one of a firm's operating department or in Its inventory control section.

The purchasing dept. is usually notified of the need by one of the three basic methods

- a. standard purchase requisition (SPR)
- b. material requirements planning (MRP)
- c. bill of materials (BOM)

a) Standard Purchase Requisition:

- It is an internal document numbers serially for requests originating in the operating Departments,
- A PR is used for materials that have to be ordered from suppliers
- The items included could be:
 - Material name/code identification
 - The amount needed
 - Desired delivery date
- PR formats vary widely because each company designs its format to simplify its own communication problem. A typical PR is given below.

Date _____ Sl.NO _____
 Date by with material required _____
 Dept. _____

Item No	Description	Qty.

 Inventor

 Authorized by

- The user dept. generally makes a minimum of two copies-one copy is sent to purchasing, the other is retained in the using dept.'s file.

Note. Stores requisition: it is used to ask for materials that are in regular use in the plant and carried out as the stock. This requisition goes directly to the stores dept. and the requirements are supplied form there.

For items of a repetitive nature and for those for which purchases are normally made to replenish stocks, a traveling requisition is used.

b) Material requirement Plan (MRP)

MRP is a technique for determining the quantity and timing for the requisition of dependent dd items.

c) Bill of Materials (BOM)

- It is a complete list of all items incorporated in to a finished product with specifications And quantity required of each item of materials.

2. Verification of Purchase Requisition

It involves the purchasing dept. responsibility for

- Checking the document for accuracy and completeness
- Determining that the need has adequately defined
- Ensuring that the appropriate method of description has been used to guarantee a satisfactory purchase for the user and at the same time all possible latitude in the selection of the supplier.

3. Request for Quotation (Bids, price quotation)

- Requests for quotation is a process of initiating potential suppliers that are willing to Compete to supply the required material
- The request depends on the type of materials b/se some materials need “Request for quotation” and others not.
- Competitive bidding is dictated by five criteria.
 - a) the dollar value of the specific purchase must be large enough to justify expense to both buyer and seller
 - b) The market must consist of an adequate No. of sellers.
 - c) The sellers that make up the market must be technically qualified and willing to compete
 - d) The specification/description is clear to both the buyer and seller
 - e) The time available must be sufficient for using competitive bidding
- Competitive bidding should not be used:
 - For situations in -able it is impossible to estimate costs with a high degree of certainty.
 - Situations in the price are not the only importance variable. Example, quality, schedule and service may well negotiable variables of equal importance
 - Repetitive and routine purchase
 - Items of low value
 - Single/few suppliers

Invitation for bids (IFB) or request for proposal (RFP) or Request for quotation (REQ) includes:

- Purchase description /specification
- Delivery schedule (timing and mode)
- Special terms/conditions
- Eligibility of suppliers
- Bid security
 - └ Bid bond
 - └ Performance bond
- Any amendments
- Address for further information
- Purchasing company
- Term of payment
- Last date of submitting bids
- Time, date and place of opening the bid

4. Evaluation and Selection of suppliers

- There are two primary supply sources: internal and external. The internal source is the company itself and the external sources are the outside suppliers and the market place. Outside suppliers are “shops” of the purchasing dept. Thus, when evaluating and selecting a supplier, a buyer should try to find one who would meet the needs of the quality, quantity and delivery time (purchase description and specification) at lower cost

5. Insurance of purchase order (PO)

- Once a supplier has been selected, the purchasing dept. prepares and issues a serially Numbered purchase order,
- Purchase order (PO) is the instrument by which goods are procured to fill a requirement. It expresses is specific language the agreement b/n the buyer and the vendor. Once accepted, it has the legal force if a binding contract.
- The essential information in every purchase order includes.

• Name and address of purchasing company • Identifying order number • Date, number and address of the vendor • General instructions • Delivery date required • Shipping instructions	• Descriptions of materials ordered and the quantity • Price and discounts • Terms and conditions (also called boiler plate) • Signature
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- Most companies prepare PO on multipart forms visually of copies these multipart forms Provide enough copies of the order to satisfy both internal and external common needs.

- Copy 1 and copy 2 (the acknowledgement copy) are sent to the supplier
- Copy 3 informs the acct dept. of the purchase. It is used in accounting in checking and issuing payment for the seller's invoice
- Copy 4 advises the stores dept. that it can expect to receive shipment of the order on a particular date.
- Copy 5 informs the user of the details of the order so he/she can plan the work and budget accordingly
- Copy 6 remains in the purchasing dept. open order file and is used often for the purposes of order follow-up and expediting
- Copy 7 becomes the buyer's working document and is filled in purchasing's open order file.

6. Follow-up and Expediting

- The objective of follow up is to see that the right quantity of materials is received at the right place. This means ensuring that
 - Quotations are received on time
 - Replies are received on time from suppliers or other dyts
 - The supplier(s) acknowledge the order and accept the delivery schedule given
 - Materials are received according to the delivery schedule
- We should have the systems which show that, the date at which delivery check should be made.
 - a) vendor files and review: This is filling orders concerting general correspondence, delivery schedules, supply progress sheets, follow-up letter in respect of the supplier concerned

Suppliers	Delivery schedule	Delivery made
A=		
B=		
C=		

If the delivery of a supplier is behind schedule or a relay regarding any point is not given:

- supplier can be contacted by personally, by telephone, letter

A remainder will be sent indicating the quantity received in the previous month, quantity outstanding, the quantity now urgently required, fresh orders placed

b) Signal method

- uses for single item orders
- The signals are placed against the "date scale" on stock control card, corresponding to The period on which supplier will be expected to arrive. The scale may be divided into four week1, 2, 3, and 4 or into dates 1, 8, 15, 22 and 29 for each month.
- If the material arrive in time, the signal is removed if does not a remainder will be sent to the supplier and the signal will be removed further to coincide with the next anticipated arrival date and so on until all the materials are received.

c) Purchase record card and review. A card which shows the total no. of items covered

By the PR, details of PO and details of purchase receipts.

P. Reg. No 8001

No of items on PR _____

Date _____

No of items not covered by orders _____

Date	Order No	Supplier	Delivery Date	No. of items	Delivery details	
					Receive	Balance

The cards should be reviewed once a week /fort might and take action where necessary.

d) Method

- Useful in cases of multi-item purchase requisition files. In the filer all maters including

the order nos. which is to be followed up will be noted down.

Expediting: It is speeding up or accelerating the receipt of the item before the agreed upon time.

7. Receipt and Inspection of Orders

This procedure involves the following activities:

- Unpacking and checking the materials-compare for the right materials (packaging ship with purchase order)
Package slips (labeling) - itemizes and describes the contents of the shipment
- Completing the receiving report and distribute to
Copy 1 purchasing dept in closing/ updating its working files of purchase order
Copy 2 Acct dept for effective payment
Copy 3 User dept. as notification that the material has arrived
Copy 4 Receiving dept. file

Receiving department: Usually an adjunct of the store dept. which may or may not be part of the purchasing dept its functions are

- Receive incoming goods
- Sizing the delivery notice presented by the carrier
- Identify and record all incoming materials
- Report their receipt to the purchasing dept.
- Make prompt disposition of the goods to the appropriate dept.

Inspection:

- Whenever it is necessary to take technical inspection, we may make sample/all inspection. This depends on the nature of material and/or the description of those materials. Based on technical report, three possible courses of actions would be:
 - return the supplier
 - keep some of the more acceptable and return the rest

Checking of invoices and Bill payment

A simultaneous check and review of purchase order the receiving report and the invoices.

- By checking the receipt report against the purchase order the purchaser determines whether the quantity and type of ordered in fact received.
- Comparing the invoice with the purchase order and receiving report the firm verifies that the supplier's bill is correctly priced and that it covers the proper quantity of acceptable material

Note. Theoretically, the purchasing dept's job is completed when the material covered by the purchase order has been received. Thus invoice auditing can be handled by accounting personnel.

8. Closing the order

Closing the order simply entails a consolidation of all documents and correspondence relevant to the order.

The completed order is filed in the close order file.

- In most forms, a completed order consists of:

- | | |
|--|---------------------|
| ➤ The purchase requisition (PR) | ➤ Receiving report |
| ➤ Copy of the purchase order (PO) | ➤ Inspection report |
| ➤ Acknowledgment | |
| ➤ Any notes and any notes/correspondence inventorying to the order | |

- The completed order file, thus constitutes records of all activities encompassing the total Purchasing cycle.

3.3. The 5R principles of purchasing

Dear students, in the previous sub section you have seen the policies and procedure of purchasing. Also steps in purchasing procedure, and how it works in different firms.

Dear student in depth explanation with practical cases about the 5R principles of purchasing will be discussed and given as follow.

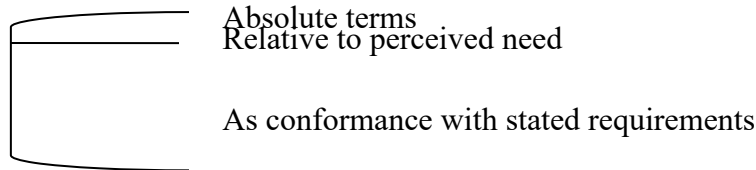
There are five basic principles of purchasing which are well enumerated as: Buying of materials in right quantity of right quality, at right price, from a right source and at right time.

Right quality

- Basic concept of quality
- Factors determining quality level
- Methods of purchase description

Basic understandings:

Basically quality can be defined in three ways:



a) Absolute view

- Quality is a function of excellence, intrinsic value, or grade as determined overtime by society generally or by designated bodies in specialized fields.

Example.

- Gold-held quality, material for long
- Materials, assured by the Ethiopian standard and quality agency.

b) Relative to perceived need

- Quality is associated with the suitability of the item in satisfying the particular need/function at hand.
- It is expression of quality in business activities.

c) Conformance with sated requirements

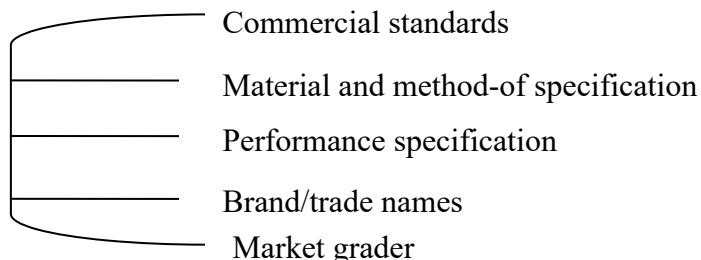
- Quality material is described as material conformance with the stated requirement,
Stated in Pr & PO

Factors Determining Quality level of purchased Materials:

Generally four factors determine the long-run quality level of a firm's purchased materials:

1. Creation of complete and appropriate specifications for quality requirements
 2. Selection of suppliers having the technical and production capabilities to the desired quality
 3. Development of a realistic understanding with suppliers of quality requirements
 4. Monitoring of suppliers' quality least performance, purchasing dept. is directly responsible for factors 2 and 3, and it should play a strong competitive role in the first and fourth factors.
- Quality Description and specification series many purposes like
 - To communicate to the buyer in the purchasing dept. What to buy
 - Communicate prospective suppliers what is required
 - Serve as the heart of the resulting PO
 - Establish the standard against which inspections, tests are made

Methods of purchase Descriptions



Commercial Standards

- A standard is uniform identification that is agree
- Commercial standards are complete description of standardized materials
- Commercial standards are applicable to raw materials, fabricated materials, parts and components, and subassemblies

Advantages

- Cheaper than those purchased without commercial standards
- Fit to standard application irrespective of where they are made
- Avoids misunderstanding

Disadvantages

- Inspection cost is high-b/se purchase materials should be checked against commercial standard
- Not items/materials in industry are covered by commercial standards
- ii) Material and methods of the buyer the supplier by the specific materials to be
Used and how they are to be processed type of technology needed
- the buyer assume full responsibility for product performance
- The buyer assumes that his/her own organization has the latest knowledge concerning materials, techniques, and manufacturing methods for the item being purchased.
- Industry using the methods
 - Large steel buyers
 - Chemical and drug buyers

Advantage

- widest competition is possible
- good pricing is assured

Disadvantage

- it puts greater responsibility on the buyer
- it can deny a co the latest advancements in both technological developments and manufacturing prepare and to inspect
- iii) Performance specifications
- Theoretically it is the perfect method of describing a requirement
- It describes an item/material, and quantity where possible, what the item is required to perform
- **Example:** A performance specifications for wire may include:
 - With stand a given temperature
 - Have a designed resistance to abrasion
 - Have a given conductive capability

Space craft, electronics, aircraft, and automobile co frequently this method

Industry uses performance specifications extensively to buy expensive, complicated machines and machine tools.

Advantage

- Ease of preparing specifications
- Assurance of obtaining the performance required
- Inclusion of applicable new developments

Disadvantage

- Poor supplier selection: because the supplier assumes the entire responsibility for designing and making the product, quality is entirely in its hands. If the supplier is not capable, it cannot apply the most advanced technical and manufacturing knowledge

i) Brand/trade names

A brand represents the manufacturer's pledge that the quality of the product will be consistent from one purchase to the next.

A buyer can be certain that a reputable manufacturer will strive to keep to keep this pledge.

Advantage

- Simple to describe on PO
- Save purchasing time and reduce purchasing expense.

Disadvantage

- Higher price
 - Lack of consistent quality unless the manufacturer has a good production and quality control mechanisms
- ii) Market grades

Grading is a method of determining quality

3.3.2. Right price

- The right price is the one which brings the ultimate value. It needs to be the lowest price

Four factors influence "right price determination"

1. Demand and supply of materials
2. competitive factors
3. cost of production and profit objective
4. discounts/reductions of purchase price

Four commonly used kinds of discounts are:

a) Cash discount

- price reductions for the prompt payment of debts
- when such discounts are given, they are offered as a percentage of the net invoice price

Example: 2/10, n/30 → a discount of 2% can be given if the invoice is paid within 10 days.

b) Quantity discount

Price reductions given to a buyer for purchasing increasingly larger quantities of materials.

- They are normally offered under one of three purchasing arrangements:

- For purchasing a specific quantity of items of one time
- For purchasing a specified birr total of any No. of d/nt items at one time.
- For purchasing a specified birr total of any No. of items over an agreed-upon time period.

c) Trade discounts: These discounts are reductions from list price allowed various classes of buyers and distributors to compensate for performing certain marketing functions to the original seller (manufacturer) of the product.

Trade discounts are frequently structured as a sequence of individual discounts (e.g. 25, 10, 5%) → series discounts

Examples

Price = birr 800

Discounts	25%	promotion
	10%	market research
	5%	installation

→ Discounts

1. $0.25 \times 800 = 200$ (for promotion)
2. $0.10 \times 600 = 60$ (for mkt research)
3. $0.05 \times 540 = 27$

Total discounts 287

Purchase price = $800 - 287 = 513$ Birr

Note: Seasonal discounts

- Price reductions based on the seasonal nature of some products.

3.3.3. Right time

- The timing of purchase varies according to

- i) Market conditions – stable and fluctuate market

ii) Current requirements

iii) Types of materials

- In any case, the basic types of purchases are:

1. Hand to mouth purchasing

- It is a practice of buying materials to satisfy current operating requirements in quantities smaller than those normally considered economical

Seasons for exercising this method are:

- When there is no enough cash-flow in the orgn.
- When there is too much inventory and insufficient storage space
- When a drop-in prices is expected in the future
- When a design change is expected in the future

Advantage

- it saves time when prices are dropping
- it prevents inventory losses that result due to technological change'
- it provides the firm required additional cash for operating purposes

Disadvantage

- it leads to higher buying and administrative expenses
- risk of running out of stock

2. Speculative purchasing

It involves purchasing in excess of normal requirements believing that the price rises.

3. Forward purchasing

It involves purchasing in excess of current requirements but taking into account the operation demand of materials.

Note: The distinction b/n forward purchasing and operator purchasing lies is the reason for making the advance purchase (excess purchaser).

- Forward purchasing is done because operational considerations or supply conditions suggest the need for inventory reserve.
- Speculative purchasing is conducted with the hope of profiting from price changes.

Advantage

- Useful of material will not be available in the market.
- Useful if price is expected to increase in the future.
- Helps to obtain quantity discounts.
- Reduces the risk of stock outs.

Disadvantage

- Price risk is involved, it may fall, Carrying cost increases.

d) Requirement purchasing

- Purchasing in excess of a hand-to-mouth quantity.
- Most common method of purchasing to satisfy short range requirements.

3.3.4. Right quantity

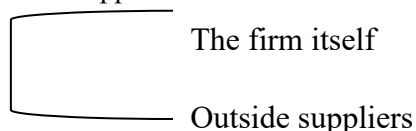
- In the case of recurring items the right quantity is the quantity that may be purchased at a minimum cost and which obviates shortage

Methods – EOQ, QD, NLP

- It the case of special purchases, if is just the quantity required for carrying out the job

3.3.5. Right Source: Supplier evaluation & selection

- A firm has two categories of suppliers



In the actual process of source selection, there are four stages

I. Survey Stage

- All possible sources are explored to obtain infn. About available suppliers; searching for all likely suppliers
- Potential sources to a buyer in establishing a list of potential suppliers include
 - Supplier information file: purchasing dept should keep supplier infn. About:
 - name of each supplier
 - list of materials available from each supplier
 - suppliers delivery history
 - The suppliers quality record supplier information files are important b/c many purchasing operations are repetitive.
 - Trade journals
 - Yellow pages
 - Trade exhibits
 - Company personnel
 - Other purchasing depts.
 - Personal contacts

II. Inquiry Stage

- Analysis of qualification and advantage of potential sources
- Inquiry stage involves prequalification of potential sources which narrows the field from possible sources to acceptable sources.

Factors to be considered:

- Location: if a supplier is located very far from the buying firm, there are greater risks of interruptions because of:
 - accidents/damage in transit
 - delay in transportation and other problems
- services, which include
 - arrangement of transport, insurance,
 - after sale service- installation, maintenance, repair warranty
 - discounts
 - convenient packaging
 - on time delivery
 - purchase reforms

III. Selection and Negotiation

- leads to the issuance of PO
- Both subjective (qualitative) approach and quantitative approach can be used

IV. Experience Stage

This stage involves:

- follow up to ensure that the supplier(s) meet the terms and conditions of the contract
- rating and evaluating supplier performance

Quantitative Supplier selection and Evaluation Methods

Dear student selection and evaluation of supplier is one of the most important tools for purchasing management.

The two commonly used methods of selecting and evaluating suppliers are

1. The weighted point method
2. The cost-ratio method

1. The weighted point Method

- The method provides qualifying criteria and corresponding relative points (weights) for respective factors.
- Each supplier is rated on the criteria and the supplier(s) with the highest composite performance index becomes the most preferred supplier

Example:

Purchasing department of Arsi University often uses weighted point method to select suppliers. The department has determined the following criteria and their respective points

<u>Criteria</u>	<u>Points</u>
Quality	40
Delivery	30
Cost reduction suggestion	20
Price	10

The accompanying table shows the performance rating of three suppliers and price per unit they quoted now.

Suppliers	Quality	Delivery	Cost reduction Suggestion	Price/unit (birr)
A	90%	80%	1	40
B	80%	90%	1	50
C	70%	100%	3	60

Required Rank the suppliers and select one

Solution

Suppliers	Quantity rating	Delivery rating	CRs rating	Price/rating	Compsion rating
A	0.9x40=36	0.8x30=24	0.2x20=4	1x10=10	74
B	0.8x40=32	0.9x30=27	0.2x20=4	0.8x10=8	71
C	0.7x40=28	1x30=30	0.6x20=12	0.67x10=6.7	76.7

Ranks = 1st C 2nd A 3rd B

Decision: supplier C is selected

2. The Cost ratio method

When using this method, the buying firm identifies the additional cost it incurs when doing business with a given supplier.

These are separated as cost associated with the quality, service and price elements of supplier performance.

- Each of these costs are **converted to cost ratio** which expresses the additional cost as a percent the buyer's total dollar purchase from that supplier.

- It will be totaled to get the supplier's overall additional cost ratio.

For purpose of analysis the supplier's price is the adjusted by applying its overall cost ratio. The higher is the ratio of the cost the lower is the rating and therefore lower chance for being selected.

Note:

$$1. \text{ Quality - cost ration (QCR)} = \frac{\text{Total quality cost}}{\text{Total purchase}} \times 100$$

$$2. \text{ Delivery cost ration (DCR)} = \frac{\text{Total delivery cost}}{\text{Total purchase}} \times 100$$

$$3. \text{ Price ratio} = \frac{\text{Total lowest price}}{\text{Actual price}} \times 100$$

$$4. \text{ Service cost ratio (SCR)}$$

In this case we need to establish a norm or standard service requirements and then we can evaluate the supplier to see whether they fall above or below the standard. Finally we should relate to the quoted price.

Example: Suppose you are the purchaser of a company and you want to select among supplies A, B, C and D based on the performance of the suppliers during the year just ended and forward your proposal to the higher mgt based cost ratio method. The records were given below:

Deliver costs (in birr)

	A	B	C	D
Telephone call	400	100	200	300
Telegrams	475	275	425	200
Expediting	875	300	975	750
Premium shipment	750	225	900	950
Miscellaneous	500	300	700	600

Quality costs (in birr)

	A	B	C	D
Visit	300	500	400	200
Sample approval	400	700	800	600
Incoming inspection	100	275	250	225
Re-work	0	425	450	400
Reject	200	1500	975	675
Other costs	0	200	325	700

Service rating (service in points)

Factors	Max pt	A	B	C	D
Financial stability	20	20	18	15	15
Filed service	25	20	21	15	12
Labour relation	10	8	10	10	8
Geographic location	15	10	15	6	0
Flexibility	5	5	5	3	5
Expansion capacity	10	10	8	7	9
Warranty	10	8	10	8	0
Miscellaneous	5	3	4	0	0

- Assume that the service rating is 60 points and the maximum value of service package is 20% of price.
- The total value of shipments (purchase) and the quoted prices are given as follows.

Supplier	Total value purchase(Birr)	Quoted price
A	100,000	85
B	120,000	86
C	80,000	82
D	140,000	83

Required: which supplier is the best based on the cost ratio-method

Solution

Step1. Summary of the costs

Suppliers	Delivery(Birr)	Quantity (Birr)	Service
A	3000	1000	84
B	1200	3600	91
C	3200	3200	64
D	2800	2800	49

Step 2. Delivery Cost ratio

<u>Supplier</u>	<u>DCR</u>
A	$\frac{3,000}{100,000} \times 100 = 3\%$
B	$\frac{1,200}{120,000} \times 100 = 1\%$
C	$\frac{3,200}{80,000} \times 100 = 4\%$
D	$\frac{2,800}{140,000} \times 100 = 2\%$

Step 3. Quality cost ration:

<u>Supplier</u>	<u>DQR</u>
A	$\frac{1,000}{100,000} \times 100 = 1\%$
B	$\frac{3600}{120,000} \times 100 = 3\%$
C	$\frac{3,200}{80,000} \times 100 = 4\%$
D	$\frac{2,800}{140,000} \times 100 = 2\%$

Step 4. Service cost rating - SCR

<u>Supplier</u>	<u>SCR</u>
A	$\left(\frac{84}{60} \times 100 \right) - 100 = 40\% \times 0.2 = 8\%$
B	$\left(\frac{91}{60} \times 100 \right) - 100 = 51\% \times 0.2 = 10.3\%$
C	$\left(\frac{64}{60} \times 100 \right) - 100 = 6.67\% \times 0.2 = 1.33\%$
D	$\left(\frac{49}{60} \times 100 \right) - 100 = -18.3\% \times 0.2 = -3.67\%$

Note: 1st determine the percentage by w/h the rated supplier is above or below the acceptable norm.

2nd Apply this percentage to the total value of the service package to determine SCR.

Step 5. Cost ratio summary

Supplier	DCR	QCR	SCR	Total
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A	3	1	-8	-4
B	1	3	-10.3	-6.3
C	4	4	-1.33	6.67
D	2	2	+3.67	7.67

Step 6. Adjust to the quoted price

Supplier	Quoted price	Incremented/decrease price	Net (1±2) adjusted price
A	85	$85 \times 0.04 = 3.4$	81.6
B	86	$86 \times 0.063 = 5.42$	80.58
C	82	$82 \times 0.0667 = 5.47$	76.53
D	83	$83 \times 0.0767 = 6.37$	89.37

Rank 1st - C
2nd - B
3rd - A
4th - D

Factors influencing make-or-buy decisions

Two factors stand out above all other when considering the make or buy decisions:

Cost and availability of production capacity.

When obsolesces makes machines worthless or substantially reduce their value

- Desire to maintain a stable work force (in periods of rising seller)
- when the demand for the product is fluctuating

Considerations which favor making

- when the cost to make is substantially lower or less than the cost to buy
- when the suppliers are unable to meet specification in terms of quality and performance
- when the company has idle capacity like idle space, skilled human resource, equipment
- need to exert direct control over production and/or quality
- design secrecy required trade secrets,
- materials in if and the process involved
- when the co experience is well suited to

Consideration which favor buying

- when the cost to buy is substantially lower or less than the cost to make
- suppliers research and specialized know-how
- small volume requirements
- limited production facilities
- desire to maintain a multiple-source policy
- when other companies hold trade secrets or patents on a required material so that it is not possible to make it

Quantitative Analysis of Make-or Buy decision

Example 1: ABC Machine Company produces parts that are shipped nationwide. It has an opportunity to produce plastic packaging cases which are currently purchased at 8 Birr each. Annual demand for the product depends largely on economic conditions and this has been estimated at 37,500.

If the company producer the cases itself, it must renovate an existing work area and purchase a molding machine which will result in annual fixed costs of Birr 8,000. Variable costs for labor, materials and per head are estimated as Birr 6 per case.

Required

- Should the company make or buy the cases?
- At what volume of production is it more profitable to produce in-house rather than purchase from an outside supplier?

Value analysis

Value analysis is concerned with scrutiny of the design function and cost of any product/material/service with the object of reducing by modification of design material specification.

- More efficient process.
- Change in source of supply.
- Elimination of an item.
- Substituting an item in the related item without sacrificing reliability and quantity.

-The purpose of value analysis is to bring together the combined talent of purchasers, engineers, and other operating personnel to review the components of material usual in making the product.

Value analysis /value engineering Tools

- The basic conceptual tools in the operation of VA value analysis are

1. Design analysis of the required product, part or material

- Design analysis involves a methodical step-by-step study of all phases of the design of a given item in relation to the function it performs.
- Decomposing an item into its parts so as to see and examine each part in relation to their function avoids or eliminates redundancy or idle ones.
- Analysis of each component/part in this fashion attempts to answer the four basic

Questions:

- a) Can any part be eliminated without impairing the operation of the complete unit?
- b) Can the design of the part be simplified to reduce its basic cost
- c) Can the design of the part be changed to permit the use of simplified or less costly production method?

It can be less expensive but equally satisfactory materials be used in the part

Reading Assignment:

Common design analysis approaches:

- a) value analysis check list
- b) functional cost approach
- c) brainstorming
- d) use of suppliers

2. Cost analysis

- Cost analysis involves the investigation of a supplier's probable costs of producing a given product
- To get selling price of supplier
 - construct estimated elemental costs for labor, material, mfg overhead and general overhead
 - total these costs to arrive at approximate theoretical total cost incurred by an efficient producer
 - Make a reasonable margin and arrive at selling price
- An experienced analyst, with the use of wage data, material price lists and various time standards, can determine a total theoretical cost.
- Cost analysis = UA tool + negotiating tool
- Cost analysis plays two major roles
 - a) CA is conducted for currently purchased items whose cost appears excessive. In such cases, the information developed from cost analysis is used as a basis for further price negotiation with the supplier.
 - b) CA also serves as a means of locating high-cost parts which should be subjected to design analysis.